



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: PUJITHA K

Reg. No.: 192524148

Course Code: CSA0309

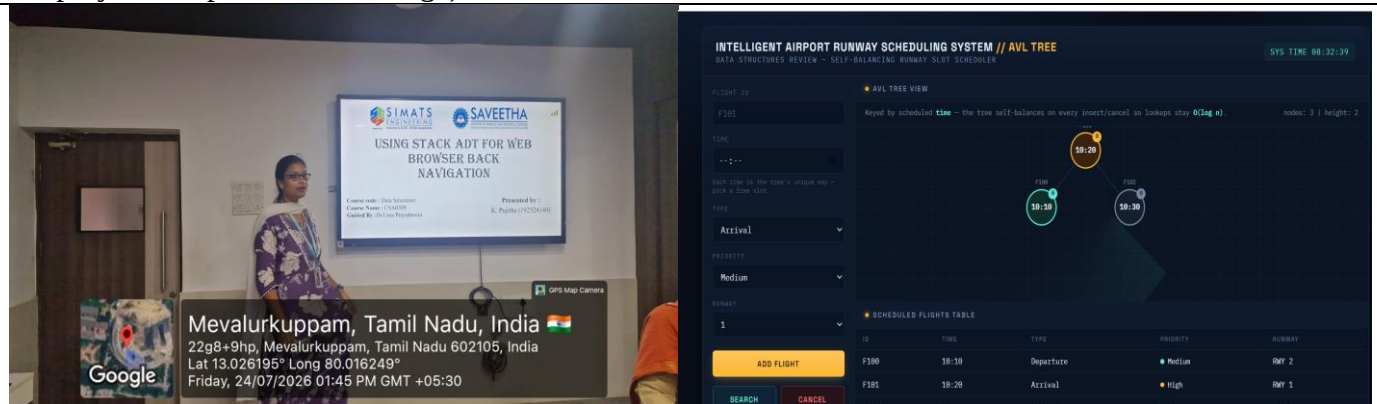
Slot: B

Course Name: DATA STRUCTURES

Course Faculty: Dr. P S UMAPRIYADARSINI

Project Title: Intelligent Airport Runway Scheduling and Conflict Management System Using AVL Trees

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

The Intelligent Airport Runway Scheduling and Conflict Management System is designed to manage aircraft landing and takeoff schedules efficiently. The system uses AVL Trees to store flight schedules according to their scheduled time. Since an AVL Tree is a self-balancing binary search tree, it provides efficient searching, insertion, and deletion of flight records. The system identifies conflicts when multiple aircraft require the same runway at overlapping time intervals. It can resolve these conflicts by changing the runway or adjusting the flight schedule based on priority and availability. This approach helps reduce aircraft waiting time, avoid runway conflicts, and improve overall runway utilization.

A system used to manage aircraft landing and takeoff times. It uses an AVL Tree to store flight schedules in an organized and balanced manner. The system checks whether two aircraft are assigned to the same runway at the same time. If a conflict occurs, the system can change the runway or adjust the flight timing. This helps reduce delays, improve runway utilization, and make airport operations more efficient.

Student Signature
PUJITHA K

Guide Signature